

L4.2 Adversarial Attacks for AD Quiz ANS

1. When the victim is a full driving stack instead of an image classifier, what replaces the small ℓ_p -norm ball as the perturbation constraint?

- A) A smaller learning rate for the attacker's optimizer, so the attack stays stealthy
- B) A requirement that the perturbation is invisible to every camera sensor
- C) A limit on how many gradient steps the attacker may run per frame
- D) Physical realizability: the attack must be a printable, projectable or placeable object

ANS: D) Physical realizability: the attack must be a printable, projectable or placeable object

2. Why is fooling an object detector in a single frame usually not enough to change what the planner sees?

- A) Planners ignore camera detections and rely on HD maps to find all objects
- B) The tracker is a temporal low-pass filter: isolated misses are bridged by prediction
- C) Detectors are re-trained online after every frame that was attacked successfully
- D) Non-maximum suppression removes every adversarial box before the tracking step

ANS: B) The tracker is a temporal low-pass filter: isolated misses are bridged by prediction

3. In the lecture's three levels of attack success, what is a Level 2 (persistent perception failure)?

- A) A single frame that is misclassified, measured by the attack success rate of the model
- B) A collision or hard braking event, measured by the collision rate of the vehicle
- C) An error that survives tracking and fusion over many frames (track loss, ID switches)
- D) A change of the model weights, measured by the accuracy drop on a held-out test set

ANS: C) An error that survives tracking and fusion over many frames (track loss, ID switches)

4. Which metric belongs to Level 3 (closed-loop safety violation)?

- A) Collision rate or time to violation of a safety rule
- B) The mAP drop of the detector on a held-out data set
- C) The number of ID switches of tracks within one video
- D) The attack success rate measured on a single frame

ANS: A) Collision rate or time to violation of a safety rule

5. Toy tracker (constant-velocity Kalman filter, 10 Hz, max_age = 5 frames). What happens if detections of an object are suppressed for 2 frames versus 8 frames?

- A) 2 frames: the track is deleted; 8 frames: no visible change to the downstream modules
- B) 2 frames: no visible change; 8 frames: the track is deleted and a new ID appears
- C) Both gaps delete the track immediately and the object is lost forever afterwards
- D) Neither gap has any effect, because a Kalman filter never forgets any track at all

ANS: B) 2 frames: no visible change; 8 frames: the track is deleted and a new ID appears

6. What did Jia et al. (ICLR 2020, 'Fooling Detection Alone is Not Enough') find about attacks on tracking?

- A) Trackers are more vulnerable than detectors, so 10 % detection success is enough
- B) Detection attacks often fail to change tracking; over 98 % detection success may be needed
- C) Tracking-by-detection cannot be attacked at all in the digital domain
- D) Hungarian association can be inverted exactly by an attacker with white-box access

ANS: B) Detection attacks often fail to change tracking; over 98 % detection success may be needed

7. Why do Ma et al. (VehicleSec 2023, WIP) use dynamic adversarial patches against multi-object tracking?

- A) Dynamic patches are much cheaper to print than static patches are
- B) Dynamic patches need no knowledge of the perception pipeline at all
- C) Static patches are always detected and removed by the Kalman filter
- D) A static patch cannot adapt to the tracker's state from frame to frame

ANS: D) A static patch cannot adapt to the tracker's state from frame to frame

8. Ma et al. report 90 % and 80 % success rates on 10 BDD video clips. How should these numbers be quoted?

- A) As proof that deployed vehicles crash in 80 to 90 % of their real encounters
- B) As the success rate of a printed physical patch measured on real public roads
- C) As preliminary, white-box, digital results, not as a general risk for deployed vehicles
- D) As the average success over all tracker designs that are used in the industry

ANS: C) As preliminary, white-box, digital results, not as a general risk for deployed vehicles

9. Why can a LiDAR not tell a real reflection from an injected one in a laser spoofing attack?

- A) Its processor cannot compute the time of flight accurately enough for this
- B) It only accepts returns whose intensity is higher than a fixed threshold
- C) It computes $R = c \cdot \Delta t / 2$ and trusts any return with the right wavelength and timing
- D) It cannot measure the angle under which a return arrives at the sensor

ANS: C) It computes $R = c \cdot \Delta t / 2$ and trusts any return with the right wavelength and timing

10. Cao et al. (CCS 2019) showed that blind LiDAR spoofing is insufficient. What was the fix?

- A) Optimization with global sampling to find point patterns accepted as a vehicle
- B) Injecting many thousands of random points to saturate the object detector
- C) Physically blocking the laser beam with a large piece of black cardboard
- D) Replacing the weights of the victim's detector through an over-the-air update

ANS: A) Optimization with global sampling to find point patterns accepted as a vehicle

11. What physical invariant does the occlusion-pattern defense (CARLO, Sun et al., USENIX Security 2020) exploit?

- A) Real vehicles always reflect with a higher intensity than the ground points around them
- B) Real vehicles block the laser, so there are no returns behind them; spoofs leave it populated
- C) Real vehicles always produce exactly 70 points in a single LiDAR scan of the scene
- D) Spoofed points always arrive earlier than the returns of real objects in the scene

ANS: B) Real vehicles block the laser, so there are no returns behind them; spoofs leave it populated

12. Why does the removal attack of Cao et al. (USENIX Security 2023, 'You Can't See Me') defeat even a robustly trained detector?

- A) It adds an adversarial patch that the network is trained to misclassify
- B) It re-trains the detector on poisoned data during the next software update
- C) It deletes the target's returns at the sensor, so the network never sees the object
- D) It jams the GPS receiver so that detections cannot be localized

ANS: C) It deletes the target's returns at the sensor, so the network never sees the object

13. Tu et al. (CVPR 2020) hide a vehicle from LiDAR detectors with which type of attack?

- A) A laser that injects phantom points in front of the victim car
- B) A universal 3D adversarial mesh placed on the vehicle's roof
- C) A fabricated V2X message that carries a fake list of objects
- D) A sticker placed on the lens of the front camera of the victim

ANS: B) A universal 3D adversarial mesh placed on the vehicle's roof

14. In the SenSys 2021 attack on LiDAR segmentation (Zhu et al.), what is the purpose of the occlusion loss L_{occ} ?

- A) To keep the features of the unrelated parts of the scene unchanged
- B) To maximize the distance between predicted and target labels of points
- C) To limit the adversarial object to at most ten points in total
- D) To keep the object from being blocked, so the laser can actually hit it

ANS: D) To keep the object from being blocked, so the laser can actually hit it

15. What does the trajectory-prediction attack of Lou et al. (USENIX Security 2024) achieve?

- A) It hides every vehicle in the scene from the LiDAR object detector of the victim
- B) It disables the lateral controller of the victim vehicle during a lane change
- C) Cardboard near a parked car makes the predictor expect it to move, causing unsafe avoidance
- D) It deletes the HD map from the memory of the victim vehicle while driving

ANS: C) Cardboard near a parked car makes the predictor expect it to move, causing unsafe avoidance

16. What makes the frustum attack of Hallyburton et al. (USENIX Security 2022) effective against camera-LiDAR fusion?

- A) It spoofs points inside the camera frustum of an object, keeping both sensors consistent
- B) It spoofs points far outside any camera frustum, away from every real object nearby
- C) It blinds the camera with a bright light so that only the LiDAR input remains
- D) It needs white-box access to the weights of the camera-LiDAR fusion network

ANS: A) It spoofs points inside the camera frustum of an object, keeping both sensors consistent

17. How does MSF-ADV (Cao et al., IEEE S&P 2021) differ from the frustum attack?

- A) It attacks only the camera branch of the fusion system and never touches the LiDAR
- B) It spoofs points with a laser in exactly the same way as the frustum attack
- C) It attacks the V2X link between the vehicles rather than the on-board sensors
- D) One 3D-printed object is optimized against the camera and the LiDAR branch at once

ANS: D) One 3D-printed object is optimized against the camera and the LiDAR branch at once

18. In the dirty road patch attack on lane centering (Sato et al., USENIX Security 2021), which stage is attacked and how is harm defined?

- A) The lane-detection DNN; harm is the lateral deviation of the vehicle
- B) The lateral controller; harm is the classification accuracy of the network
- C) The GPS module; harm is the localization drift of the vehicle
- D) The planner's cost map; harm is the route length driven by the vehicle

ANS: A) The lane-detection DNN; harm is the lateral deviation of the vehicle

19. Why is Acero (Song et al., USENIX Security 2023) not stopped by defenses such as adversarial training or sensor consistency checks?

- A) It uses only realistic, legal driving maneuvers; no pixel, point or signal is modified
- B) It only works against vehicles that do not carry any perception sensors at all
- C) It modifies the neural network weights of the vehicle through a fake software update
- D) It relies on a laser attack against the front camera of the victim vehicle

ANS: A) It uses only realistic, legal driving maneuvers; no pixel, point or signal is modified

20. Why do numbers such as CARLO's 5.5 % residual attack success overstate robustness?

- A) They were measured on a data set that does not contain any vehicles at all
- B) Defenses cannot be evaluated by the success rate of an attack in any setting
- C) A residual success rate that is below 10 % does not have any statistical meaning
- D) They come from the published, non-adaptive attack; an adaptive attacker can model the check

ANS: D) They come from the published, non-adaptive attack; an adaptive attacker can model the check